

Chapter 5 Review

Learning Objectives

- Consolidate understanding of all algorithmic concepts from Chapter 5
- Apply algorithmic thinking to novel, unfamiliar problems
- Evaluate and compare different algorithmic approaches to the same problem

Engage (5-7 minutes)

Start with a “mystery algorithm” challenge: students are given input-output examples and must deduce the algorithm. This requires understanding of sequence, selection, and iteration simultaneously. Examples include a number-guessing algorithm, a pattern generator, and a data transformation process.

Explore (10-12 minutes)

Students complete a concept mapping activity linking all Chapter 5 topics: algorithms → flowcharts → pseudocode → constructs (sequence, selection, iteration) → sorting → searching → efficiency → decomposition → project. They identify connections and dependencies between concepts. Then they solve three integrated problems that require multiple concepts.

Explain (10-12 minutes)

Chapter 5 established the foundations of computational thinking. Algorithms are precise instructions; flowcharts and pseudocode are representation tools; sequence, selection, and iteration are the core constructs; sorting and searching are fundamental operations; efficiency analysis guides algorithm selection; and decomposition manages complexity. Together, these concepts enable systematic problem-solving that scales from simple tasks to complex systems.

Elaborate (8-10 minutes)

Cross-cutting review problems: design a sorting algorithm that uses binary search to find insertion positions; decompose a problem where each sub-problem requires different loop types; analyze the efficiency of a system where multiple algorithms interact. Discuss how these concepts connect to programming languages and real software development.

Evaluate (5-8 minutes)

Comprehensive assessment: a mix of short-answer (define, explain), practical (write an algorithm, create a flowchart), and analytical (compare two approaches, identify inefficiencies) questions. Students also complete a self-assessment identifying their strongest and weakest areas, and set goals for improvement. Peer teaching: students explain their weakest concept to a partner.